PREP 5 · LESSON 9

How much is useful?

Efficiency as a decimal and a percentage · Sankey diagrams

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Define power. DN 31 [1]

2 State what is meant by dissipated energy. DN 654 [1]

3 Two motors lift the same load through the same height. Motor A is quicker. Which is more powerful, and why? DN 656 [1]

Key ideas LESSON 9

Show your working

4 A motor is supplied with 800 J. It transfers 600 J usefully. Use $\text{efficiency} = \frac{\text{useful output}}{\text{total input}}$

Calculate the efficiency of the motor, as a decimal and as a percentage. [3]

5 A 40 W lamp is 30% efficient. Calculate its useful power output. [2]

6 A lamp is supplied with 100 J and transfers 20 J by light. In a Sankey diagram, the wasted arrow shows _____ (20 J / 80 J / 120 J) [1]

7 Explain why no device can have an efficiency greater than 1 (100%). [1]